

# Review Cheat Sheet: RUDP and Smart-RUDP

*Two pages to read before you go in. 9 Oct 2026.*

## Why this is a project, not a lab exercise

A lab follows a given specification. We chose a problem (build TCP's guarantees on UDP), designed several cooperating parts (13-byte header, handshake, Go-Back-N, cumulative ACKs, fast retransmit, AIMD, loss simulation), measured them (benchmark sweep, window graphs, Wireshark captures with our own dissector, Smart-RUDP vs AIMD), documented what broke and how we fixed it, and used the result in a routed network design and a working chat between two devices. We also report a negative result: Smart-RUDP did not beat AIMD.

**One line:** a lab exercise sends a datagram; we built the reliability and congestion control that sit on top of it, measured them, and used them in a real application.

## Numbers to remember

| Item                      | Value                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Header                    | 13 bytes: seq 4, ack 4, flags 1, checksum 2, length 2                                                                                    |
| Data packet on the wire   | 53 bytes (ACK or handshake: 13)                                                                                                          |
| Timeout / fast retransmit | 500 ms / 3 duplicate ACKs                                                                                                                |
| Window                    | slow start to 16, then +1 per RTT, halve on dup ACKs, reset to 1 on timeout, cap 32                                                      |
| Benchmark                 | 62 of 62 packets delivered from 0 to 40% loss                                                                                            |
| Smart-RUDP                | Q-learning, 80 states, 5 actions, 40 training episodes, 6 eval runs per path                                                             |
| Smart-RUDP result         | Experiments: +63% time on 5G-like, +10% satellite, tie on lossy. Chat (simulated): faster than AIMD on 5 of 6 held-out paths, equal on 1 |
| Network                   | 4 routers, OSPF area 0, 2 VLANs, DHCP, ACL; server LAN metric 2 via R3, branch metric 3 via R2                                           |
| Failover                  | shut R1 Gi0/2: 70 of 71 pings, rerouted via R2                                                                                           |
| Chat                      | window 8, 300 ms timeout; terminal and browser; laptop and phone over Tailscale                                                          |

## Hard questions, short answers

**Why not TCP?** TCP is in the kernel and cannot be changed. Building our own makes every mechanism visible. QUIC and HTTP/3 do the same.

**Stress test?** 48 unseen random paths x 3 seeds (loss to 30%, jitter, bottlenecks): mean 3.73 s Smart, 3.88 s fixed, 7.16 s AIMD. Smart beat AIMD on 27, lost on 5, within 5% on 16. Weak: tight queues, very high loss. A broader-trained agent was worse, so we kept the first. Live: 200 msgs at 30% loss both ways arrived in order in 4.3 s.

**Deep RL?** Yes, in the chat: `web/dqn.py`, a numpy DQN (5-32-32-5, replay buffer, target network), trained in the simulator. Mean 3.20 s vs table 3.73 s vs AIMD 7.16 s on the 48 unseen paths; faster than the table on 19, slower on 6. Seed-sensitive (1 of 4 seeds failed), so we pick by validation. Python only, simulated; the C agent is still tabular.

**TCP and UDP baselines?** Simulated on the same 48 paths: plain UDP delivers 68% (all messages on 4 of 48 paths). TCP model (selective ACK): mean 4.56 s, median 1.48 s, the best median. Deep Smart-RUDP 3.20 s mean, about even with it path by path (22 faster, 23 slower). TCP row is a model, not kernel TCP; real netem script provided, not run here.

**Why is RUDP slower than TCP?** Go-Back-N resends the whole window. A selective-ACK variant in the simulator cut AIMD from 7.16 s to 4.90 s (median 2.72 to 1.44). First item of future work.

**Reordering?** Tested in the simulator only: AIMD with Go-Back-N goes 0.82 s to 6.79 s at 20 ms jitter; learned agents cope better; SACK removes most of the damage.

**How sure are the C results?** 20 runs per mode, 95% intervals: 5G-like Smart +0.92 s (slower), satellite-like +6.3 s (slower), lossy -0.37 s (tie).

**Does Smart-RUDP beat AIMD now?** The neural agent does, in the C transport on the three emulated paths: 1.18 vs 1.38 s (5G-like), 6.63 vs 9.13 s (lossy), 13.1 vs 17.9 s (satellite-like), 95% intervals exclude zero. The table agent does not (slower on 5G-like and satellite-like). Caveats: trained in a simulator, same start window 10, AIMD cap 128 vs agents 32, more retransmissions on random loss.

**SACK in C?** Built (`make sack`, `--sack 1`): about 10x fewer retransmissions, no significant change in completion time on these paths. In the simulator it gave large gains; those paths differ.

**How do you know it works?** `make test`: wire format, end to end at 0% and 20% loss, every controller mode with and without SACK, simulator behaviour.

**Did Smart-RUDP win?** Depends on the setting. Main experiments (5G, lossy, satellite): tied on one path, slower on two (20-run intervals confirm), and we report that. Chat, retrained on simulated bursty random-loss paths: faster than AIMD on 5 of 6 held-out paths, equal on 1, because AIMD resets to 1 on every timeout. Both start from window 10; results are simulated; the agent may not shrink the window without a congestion signal (without that rule it lost).

**Why Go-Back-N?** Simpler to build and explain. It wastes bandwidth under loss; Selective Repeat with SACK is future work.

**Tests ran on one machine?** Yes, with emulated delay, loss and jitter. They do not prove internet performance and the report says so. The laptop-to-phone chat is a small real-network check.

**What does Packet Tracer prove?** The network design (OSPF, VLANs, ACL, failover). It cannot run C, so the protocol is validated separately with emulation and Wireshark.

**What if the peer disappears?** The sender keeps retransmitting on timeout. A connection abort after repeated timeouts is not built; it is the fix we would add.

**Does the checksum catch errors?** 16-bit one's-complement sum like UDP and TCP; a corrupted packet fails and is retransmitted. It cannot catch every error pattern.

**Is the loss real?** Benchmark and chat loss is simulated in the program. We also captured a real loopback run with tshark and netem loss.

**Why no handshake in the exported trace?** The export builds synthetic packets. The real tshark capture shows SYN, SYN-ACK, ACK.

**Is the Q-learning really learning?** Yes: the learning-curve figure shows it. It is small and tabular with limited training.

**Who did what?** Agree one answer as a team before you go in.

## Say the limits first

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Go-Back-N wastes bandwidth. Protocol tests are loopback only. Smart-RUDP did not beat AIMD. The agent is small. **Next:** make SACK the default, adaptive timeout in C, fairness in C, real-network tests.

## If asked: is the win real?

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Our first simulator allowed a **dup-ACK retransmission storm**; with the realistic rule the simulated gain nearly vanishes (stress mean: AIMD 7.75 s, neural 7.27 s). The evidence we rely on is the **C transport with equal window caps (AIMD capped at 32): neural still wins by 19%, 29%, 28%, all significant** (10 paired runs). Three emulated paths, not real networks. A retrained agent in the realistic simulator was stable but much worse in C, so not used.

## If asked: is it fair? does every part matter?

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Fair against itself and on short paths; **74% share against AIMD on the 120 ms path** (Jain 0.82), simulator only. Ablation: only the **no-shrink mask** matters (6.54 s without, about 3.3 s with, AIMD 7.16 s). Training is **seed-sensitive** (3.2 to 7.4 s), so we pick the seed on validation paths.